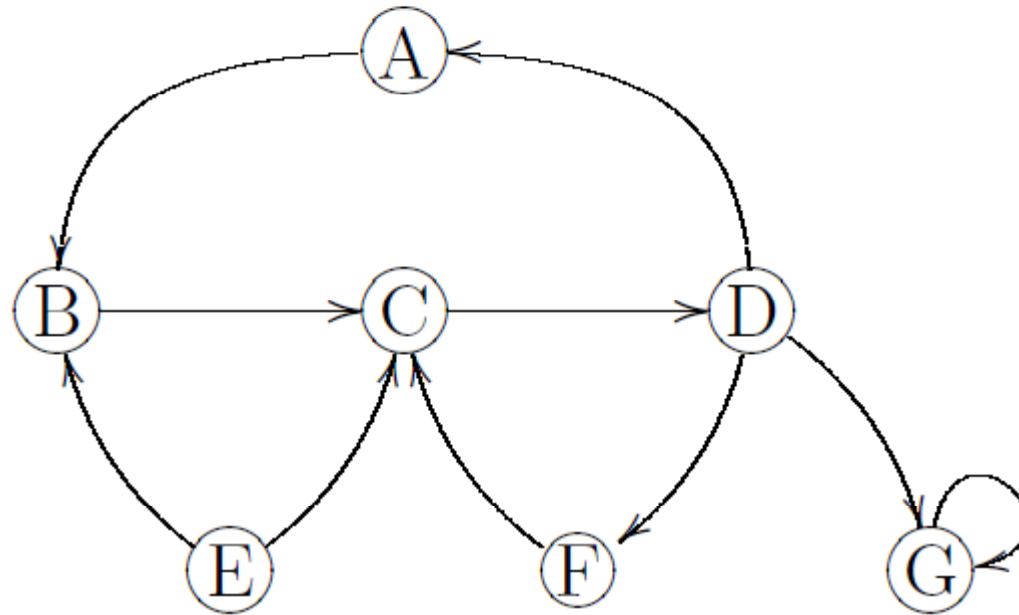


# Solution to Exercise 1

- Compute the pageranks of the nodes in the following webgraph. Use a damping factor  $\beta=0.8$ :



# Solution to Exercise 1

- The system of equations we need to solve is the following:

$$A = 0.8 * D/3 + 0.2$$

$$B = 0.8 * (A + E/2) + 0.2$$

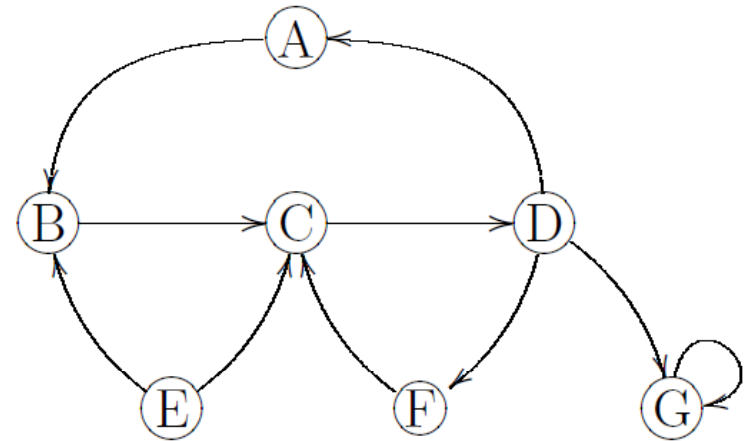
$$C = 0.8 * (B + E/2 + F) + 0.2$$

$$D = 0.8 * (C) + 0.2$$

$$E = 0.2$$

$$F = 0.8 * (D/3) + 0.2$$

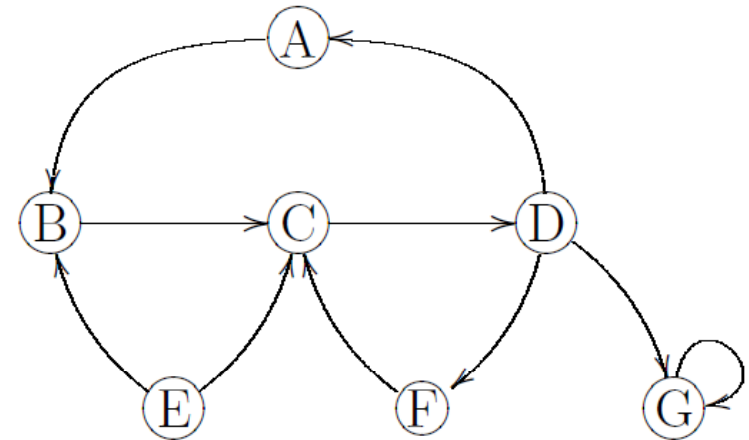
$$G = 0.8 * (D/3 + G) + 0.2$$



- We can solve it using the power iteration method, or just **Gaussian elimination**.

# Solution to Exercise 1

- Notice incidentally that the system of equalities equals the matrix formulation:



$$\begin{pmatrix} A \\ B \\ C \\ D \\ E \\ F \\ G \end{pmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1/3 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1/2 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \end{bmatrix} \begin{pmatrix} A \\ B \\ C \\ D \\ E \\ F \\ G \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 & 1/3 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1/2 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \\ D \\ E \\ F \\ G \end{pmatrix} + \begin{pmatrix} 0.2 \\ 0.2 \\ 0.2 \\ 0.2 \\ 0.2 \\ 0.2 \\ 0.2 \end{pmatrix}$$

(taking into account  
 $A+B+C+D+E+F+G = 1$ )

# Solution to Exercise 1

- The solution of the system of inequalities is as follows:

$$A = 0.52\dots$$

$$B = 0.70\dots$$

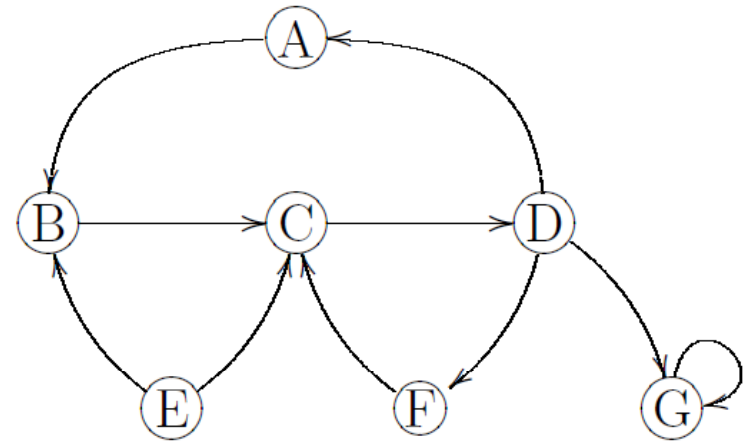
$$C = 1.25\dots$$

$$D = 1.20\dots$$

$$E = 0.20\dots$$

$$F = 0.52\dots$$

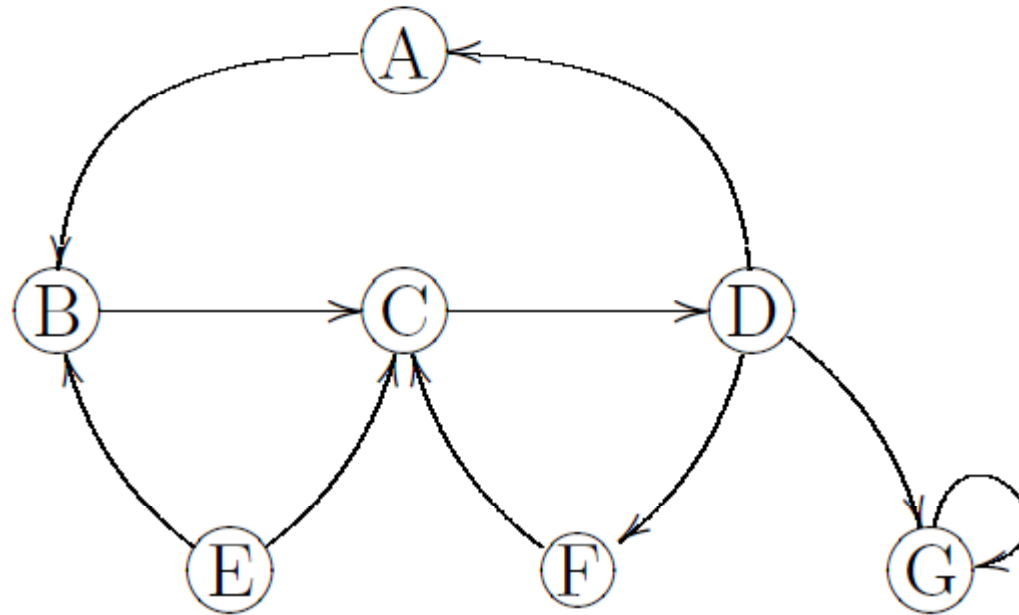
$$G = 2.60\dots$$



So, G has the highest pagerank, followed (at a large distance) by C and D.

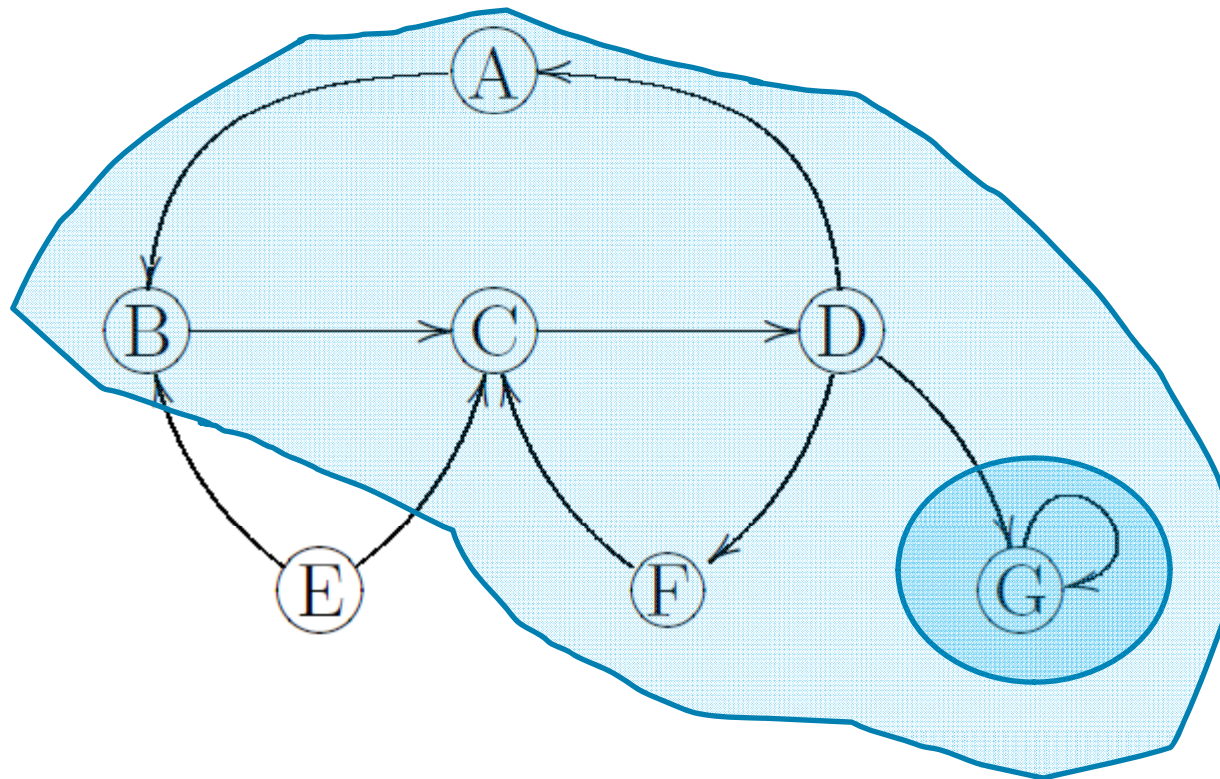
# Solution to Exercise 1

- Does the Graph have spider traps? Dead ends?



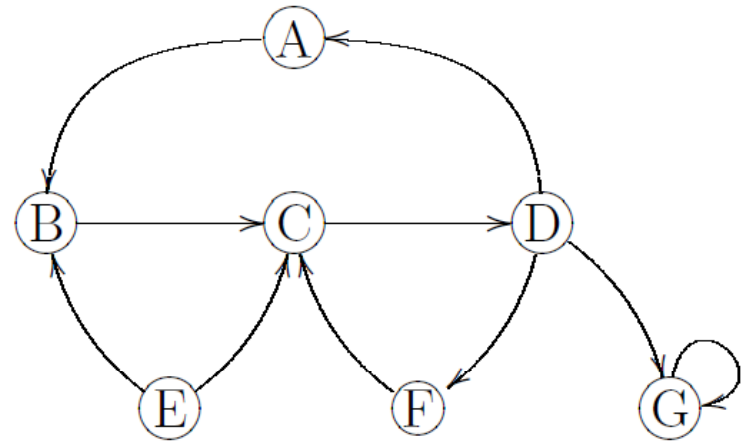
# Solution to Exercise 1

- Does the Graph have spider traps? Dead ends?
  - No dead ends
  - 2 spider traps indicated below:



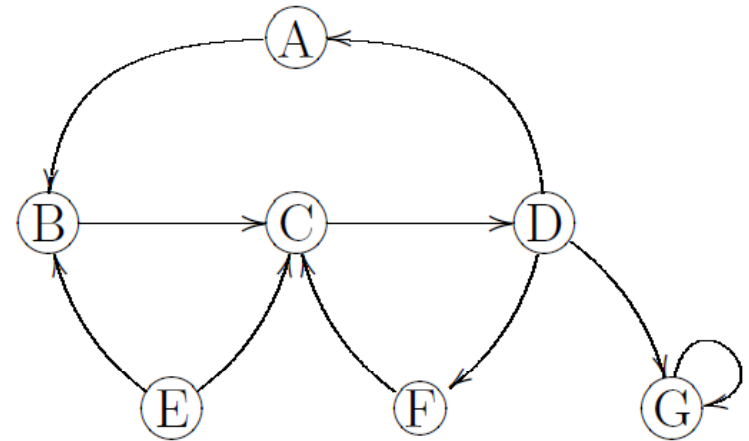
# Solution to Exercise 1

- Compute the pagerank of the nodes, but this time without a damping factor



# Solution to Exercise 1

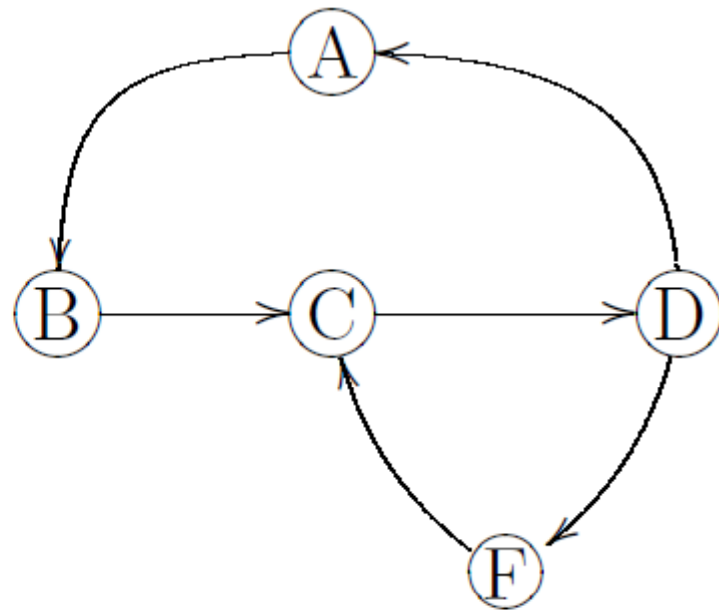
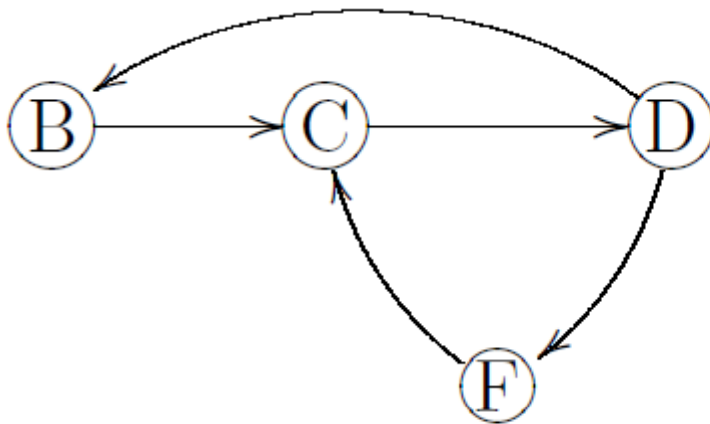
- Compute the pagerank of the nodes, but this time without a damping factor



Because of the spider trap  $\{G\}$ , all ‘importance’ will flow to the node G. Hence, in the end, all nodes will have a pagerank of 0, except for G that will have accumulated all importance.

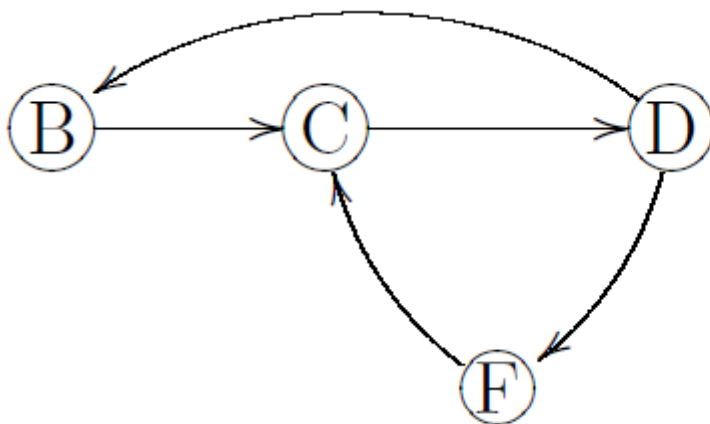
## Solution to Exercise 2

- Do the following web-graphs fulfill the aperiodicity and reducibility conditions?



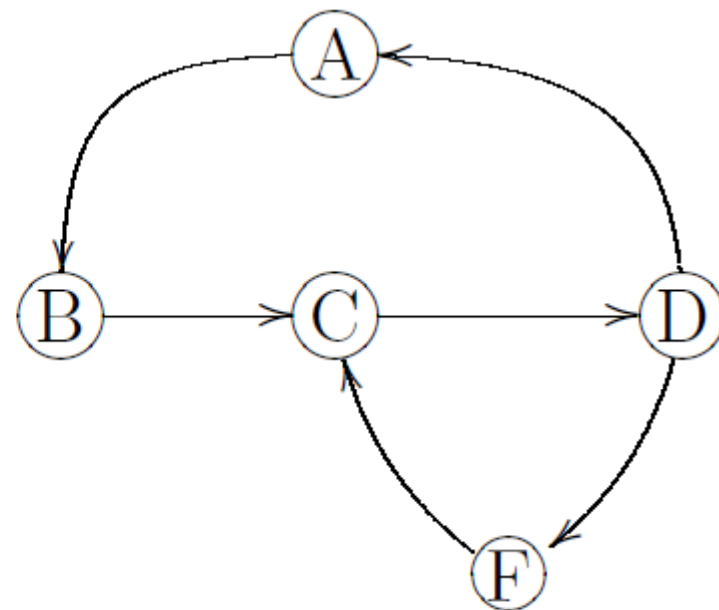
## Solution to Exercise 2

- Do the following web-graphs fulfill the aperiodicity and reducibility conditions?



Reducible

**NOT Aperiodic**; time between two visits of B is always a multiple of 3



Reducible

**Aperiodic**; for any two nodes the time between two visits can be written as  $3k + 4l$ ; hence gcd of the time between subsequent visits is 1

## Solution to Exercise 2

- What may go wrong if aperiodicity or irreducibility are not fulfilled?
- The following statement will not necessarily hold anymore: *“the stationary distribution is unique and eventually will be reached no matter what the initial probability distribution at time  $t = 0$ .”*

**NOTE: additional condition: the matrix needs to be column-stochastic! (i.e., columns need to add up to 1) Hence: teleport from dead-end with probability 1**

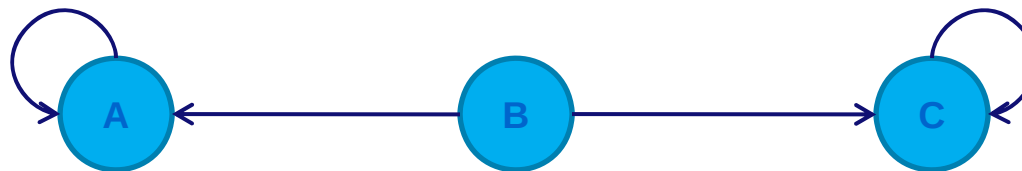
## Solution to Exercise 2

- What may go wrong if aperiodicity or irreducibility are not fulfilled?

**Irreducibility:** the solution may not be unique; i.e., there is more than one stationary distribution of the Markov process

### EXAMPLE

Not irreducible; e.g., B not reachable from A



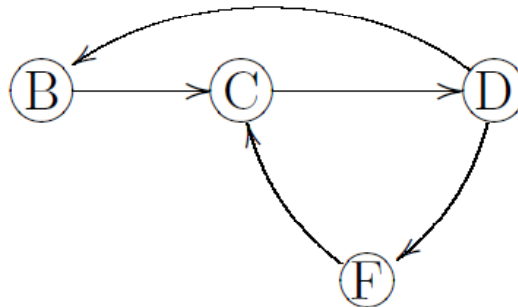
Many solutions:  $A=p$ ,  $B=0$ ,  $C=1-p$ , for all  $p$  in  $[0,1]$

## Solution to Exercise 2

- What may go wrong if aperiodicity or irreducibility are not fulfilled?

aperiodicity: the iterative procedure may not converge anymore

### EXAMPLE



#### Power iteration method

|   |     |     |     |     |
|---|-----|-----|-----|-----|
| B | 1/4 | 1/8 | 1/8 | 1/4 |
| C | 1/4 | 1/2 | 1/4 | 1/4 |
| D | 1/4 | 1/4 | 1/2 | 1/4 |
| F | 1/4 | 1/8 | 1/8 | 1/4 |

**CYCLE**

Nevertheless there is one unique solution (use Gauss):  
 $B=1/6$ ,  $C=1/3$ ,  $D=1/3$ ,  $F=1/6$

# Solution to Exercise 2

- Show that with a damping factor both conditions are always fulfilled
- Easy to see:
  - Aperiodicity: we can always 'teleport' from a node to itself = time between is 1; so not always a multiple of some  $k > 1$
  - Irreducibility: every node is reachable from every other node; we can always teleport from one node to the other